

TCC805x MCU BSP

API Specification for General Purpose Input Output

Rev. 1.00 [A]

2020-11-12

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1 INTRODUCTION

This document describes the general purpose input/output device driver (hereinafter referred to as GPIO driver) of TCC805x MCU BSP. For more detailed information of the general purpose input/output in the TCC805x MCU Sub-system, refer to Chapter 23 General Purpose I/O (GPIO) and Port Multiplexing of the *“PART 11-MICOM SUB-SYSTEM”* in the *“TCC805x Full Specification”*. [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

GPIO	General Purpose Input Output
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3 SOURCE FILES

3.1 Location of Source Files

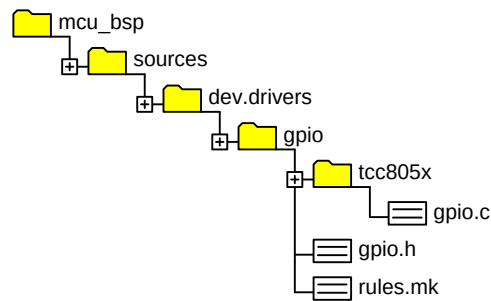


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- gpio.c
 - GPIO driver source
- gpio.h
 - GPIO driver API header
- rules.mk
 - GPIO make rule file

4 CONSTANTS

4.1 GPIO Port Configuration

Constant values for GPIO port configuration

Declaration

```
#define GPIO_SLEW_RATE_SHIFT      13
#define GPIO_SLEW_RATE_SLOW      (1 << GPIO_SLEW_RATE_SHIFT)
#define GPIO_SLEW_RATE_FAST      (0 << GPIO_SLEW_RATE_SHIFT)

#define GPIO_INPUT_TYPE_SHIFT     12
#define GPIO_INPUT_SCHMITT       (1 << GPIO_INPUT_TYPE_SHIFT)
#define GPIO_INPUT_CMOS          (0 << GPIO_INPUT_TYPE_SHIFT)

#define GPIO_INPUTBUF_SHIFT       10
#define GPIO_INPUTBUF_MASK       0x3
#define GPIO_INPUTBUF_EN         ((2 | 1) << GPIO_INPUTBUF_SHIFT)
#define GPIO_INPUTBUF_DIS        ((2 | 0) << GPIO_INPUTBUF_SHIFT)

#define GPIO_OUTPUT_SHIFT         9
#define GPIO_OUTPUT              (1 << GPIO_OUTPUT_SHIFT)
#define GPIO_INPUT               (0 << GPIO_OUTPUT_SHIFT)

#define GPIO_CD_SHIFT            6
#define GPIO_CD_MASK             0x7
#define GPIO_CD(x)               (((x) & GPIO_CD_MASK) | 0x4) << GPIO_CD_SHIFT)

#define GPIO_PULL_SHIFT          4
#define GPIO_PULL_MASK           0x3
#define GPIO_NOPULL              (0 << GPIO_PULL_SHIFT)
#define GPIO_PULLUP              (1 << GPIO_PULL_SHIFT)
#define GPIO_PULLDN              (2 << GPIO_PULL_SHIFT)

#define GPIO_FUNC_MASK           0xF
#define GPIO_FUNC(x)             ((x) & GPIO_FUNC_MASK)
```

Description

Value	Description
<i>GPIO_SLEW_RATE_SHIFT</i>	<i>Slew rate shift for GPIO_Config() 2nd parameter parsing</i>
<i>GPIO_SLEW_RATE_SLOW</i>	<i>Slew rate slow</i>
<i>GPIO_SLEW_RATE_FAST</i>	<i>Slew rate fast</i>
<i>GPIO_INPUT_TYPE_SHIFT</i>	<i>Input type shift for GPIO_Config() 2nd parameter parsing</i>
<i>GPIO_INPUT_SCHMITT</i>	<i>Input Schmitt</i>
<i>GPIO_INPUT_CMOS</i>	<i>Input CMOS</i>
<i>GPIO_INPUTBUF_SHIFT</i>	<i>Input buffer shift for GPIO_Config() 2nd parameter parsing</i>
<i>GPIO_INPUTBUF_MASK</i>	<i>Input buffer mask</i>
<i>GPIO_INPUTBUF_EN</i>	<i>Input buffer enable</i>
<i>GPIO_INPUTBUF_DIS</i>	<i>Input buffer disable</i>
<i>GPIO_OUTPUT_SHIFT</i>	<i>Output shift for GPIO_Config() 2nd parameter parsing</i>
<i>GPIO_OUTPUT</i>	<i>GPIO Output</i>
<i>GPIO_INPUT</i>	<i>GPIO Input</i>
<i>GPIO_CD_SHIFT</i>	<i>GPIO driver strength shift for GPIO_Config() 2nd parameter parsing</i>
<i>GPIO_CD_MASK</i>	<i>GPIO driver strength mask</i>
<i>GPIO_CD(x)</i>	<i>GPIO driver strength</i>
<i>GPIO_PULL_SHIFT</i>	<i>GPIO pull shift for GPIO_Config() 2nd parameter parsing</i>
<i>GPIO_PULL_MASK</i>	<i>GPIO pull mask</i>
<i>GPIO_NOPULL</i>	<i>GPIO No pull</i>
<i>GPIO_PULLUP</i>	<i>GPIO pull up</i>
<i>GPIO_PULLDN</i>	<i>GPIO pull down</i>
<i>GPIO_FUNC_MASK</i>	<i>GPIO function mask</i>
<i>GPIO_FUNC(x)</i>	<i>GPIO function</i>

Remark

None

4.2 GPIO Port & Pin Structure

Constant values for GPIO port address

Declaration

```
#define GPIO_PORT_SHIFT    5
#define GPIO_PORT_MASK    (0x1F << GPIO_PORT_SHIFT)

#define GPIO_GPA(x)        (GPIO_PORT_A | ((x) & 0x1F))
#define GPIO_GPB(x)        (GPIO_PORT_B | ((x) & 0x1F))
#define GPIO_GPC(x)        (GPIO_PORT_C | ((x) & 0x1F))
#define GPIO_GPE(x)        (GPIO_PORT_E | ((x) & 0x1F))
#define GPIO_GPG(x)        (GPIO_PORT_G | ((x) & 0x1F))
#define GPIO_GPH(x)        (GPIO_PORT_H | ((x) & 0x1F))
#define GPIO_GPK(x)        (GPIO_PORT_K | ((x) & 0x1F))
#define GPIO_GPMA(x)       (GPIO_PORT_MA | ((x) & 0x1F))
#define GPIO_GPMB(x)       (GPIO_PORT_MB | ((x) & 0x1F))
#define GPIO_GPMC(x)       (GPIO_PORT_MC | ((x) & 0x1F))
#define GPIO_GPMD(x)       (GPIO_PORT_MD | ((x) & 0x1F))
```

Description

Value	Description
GPIO_PORT_SHIFT	GPIO port shift
GPIO_PORT_MASK	GPIO port mask
GPIO_GPA(x)	GPIO A port address
GPIO_GPB(x)	GPIO B port address
GPIO_GPC(x)	GPIO C port address
GPIO_GPE(x)	GPIO E port address
GPIO_GPG(x)	GPIO G port address
GPIO_GPH(x)	GPIO H port address
GPIO_GPK(x)	GPIO K port address
GPIO_GPMA(x)	GPIO MA port address
GPIO_GPMB(x)	GPIO MB port address
GPIO_GPMC(x)	GPIO MC port address
GPIO_GPMD(x)	GPIO MD port address

Remark

None

5 APIs

5.1 GPIO_Config

GPIO port configuration user interface APIs. See the constants in Chapter 4.1 GPIO Port Configuration for supported configuration.

Declaration

```
SALRetCode_t GPIO_Config  
(  
    uint32 uiPort,  
    uint32 uiConfig  
);
```

Parameters

Value	Description
uiPort	[in] GPIO port number
uiConfig	[in] GPIO port configuration

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

5.2 GPIO_Get

Gets the value of the GPIO port corresponding to the input parameter

Declaration

```
uint8 GPIO_Get
(
    uint32 uiPort
);
```

Parameters

Value	Description
uiPort	[in] GPIO port number

Return

Value	Description
0	GPIO port value low
1	GPIO port value high

Remark

None

5.3 GPIO_Set

Sets the 2nd input parameter value to the GPIO port corresponding to the 1st input parameter

Declaration

```
SALRetCode_t GPIO_Set
(
    uint32 uiPort,
    uint32 uiData
);
```

Parameters

Value	Description
uiPort	GPIO port number
uiData	GPIO port value

Return

<i>Value</i>	<i>Description</i>
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

5.4 GPIO_ToNum

Returns the GPIO port pin number.

Declaration

```
sint32 GPIO_ToNum
(
    uint32 uiPort
);
```

Parameters

<i>Value</i>	<i>Description</i>
<i>uiPort</i>	<i>GPIO port number</i>

Return

<i>Value</i>	<i>Description</i>
<i>Greater than 0</i>	<i>Port pin number</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

6 HOW TO USE GPIO DRIVER

6.1 Example of GPIO Driver APIs

```
int main()
{
    SALRetCode_t  ret = SAL_RET_FAILED;
    uint32        value;
    uint32        pin_num;

    ret = GPIO_Config(GPIO_GPMA(27UL),           // target port : GPMA Port 27th pin
                      ((uint32)GPIO_FUNC((uint32)10UL) | // configuration : Function 10,
                      (uint32)GPIO_OUTPUT |           // GPIO output mode
                      GPIO_PULLDN |                  // GPIO pull down
                      GPIO_CD((uint32)0x3UL));        // GPIO driver strength

    Value         = GPIO_Get(GPIO_GPK(5));           // return the value of GPIO_K port 5 pin
    ret           = GPIO_Set(GPIO_GPK(5), 1UL);      // set 1UL value on GPIO_K port 5 pin
    pin_num       = GPIO_ToNum(GPIO_GPB(3));         // return the value of GPIO_B port 3 pin
    number

    Return ret;
}
```

7 REFERENCES

- [1] TCC805x Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

8 REVISION HISTORY

Rev. 1.00: 2020-11-12

- First version release

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